AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 12 · STUDENT LESSON

Nearest Neighbors

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Measure the distance. Order the IDs. Count the votes.

Explain nearest-neighbour predictions under explicit pool, distance, k and tie choices.

Keep one habit: Complete training pool → ranking script → labelled length plot → validation map → optional scaling contrast → independent query.

MAIN LESSON · SECTION 1

Which stored examples are nearest under the stated rule?

> Distance, k and tie policy together determine this classroom prediction.

Learning goals

I can compute absolute **distance**s and rank **training neighbour**s.

I can apply a declared **neighbour** count and **distance** tie rule.

I can evaluate **neighbour prediction**s without using validation labels as **neighbour**s.

Retrieve the stored-tree distinction

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Dataset Cards. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Label Quality. Give an example and a mistake that the idea helps you avoid.
- What does a fitted depth-one tree store, and what does it do with a new example?

End goal

By the end, I can explain nearest **neighbor**s using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

k-nearest **neighbor**s predicts from nearby labeled examples under a chosen feature scale, **distance** rule, and value of k.



LEARNING TARGET **Training** lengths 2(H),6(S),10(H); query=8. Compute **distance**s. Explain the tie and why combining length in mm with width in cm changes a two-feature **distance**.

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

Read the complete training pool before measuring distance

> Use only the declared feature; IDs audit rows and break ties.

Nearest-neighbour idea and fictional source

A nearest-**neighbour** classifier predicts using nearby **training** examples rather than following a fitted tree.

This week we store labelled examples, measure how different their feature values are from a new query, and let selected **neighbour**s **vote**.

Nearby means close under our chosen feature and **distance**, not necessarily similar in every real-world way.

Our new fictional dataset has a **distance** reading in centimetres and a separately reviewed lane label. Each row comes from a distinct trip, and validation trips are separate. Only the numeric reading measures similarity.

IDs identify rows and break **distance** ties; they are not numeric features. These invented readings are not a robot safety system.

All five stored training rows

ID	Reading in cm	Label
A	2	blocked
B	4	clear
C	8	clear
D	10	blocked
E	12	clear

Distance, k and tie contract

For this one-dimensional feature, **distance** is abs(**training** reading minus query reading). abs removes the sign. Set k=3 and give each selected **neighbour** one **vote**.

Order equal **distance**s by ID alphabetically, declared before scoring. In this binary, three-**vote** exercise the label **vote** cannot tie, but the **distance**s can.

▼ Earlier wording, alternatives and picture transcripts

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



MYSTERY The team must use nearest **neighbor**s to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Sort distance and ID, then select exactly k neighbours

> A tie for distance differs from a tie in the label vote.

Complete query3 program

```
train = [("A", 2, "blocked"), ("B", 4, "clear"),
         ("C", 8, "clear"), ("D", 10, "blocked"),
         ("E", 12, "clear")]
query = 3
ranked = sorted((abs(x - query), item_id, label)
                for item_id, x, label in train)
near = ranked[:3]
labels = [row[2] for row in near]
prediction = "clear"
if labels.count("blocked") > labels.count("clear"):
    prediction = "blocked"
print([row[1] for row in near])
print(prediction)
```

Trace distances and the three selected IDs

For query 3, **distance**s to A through E are 1,1,5,7,9 cm. sorted orders tuples by **distance** first, then the unique ID. ranked[:3] takes the first three rows.

The **output** is ['A', 'B', 'C'] then clear: one blocked **vote** and two clear **vote**s.

Compare the declared k=1 alternative

If k were 1 under the same **distance** tie rule, A would win and predict blocked. Changing k can change a **prediction**.

We declared k=3 for this exercise; do not choose k after seeing the desired answer and pretend the method was unchanged.

State unvalidated input assumptions

The code assumes unique IDs, numeric readings, only clear or blocked labels and at least three **training** rows. It does not validate those conditions.

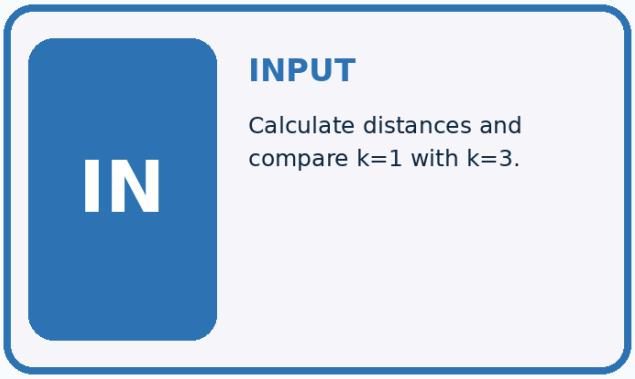
A larger k than the stored row count would not produce the promised number of **vote**s. Keep unresolved review labels under a separate documented policy.

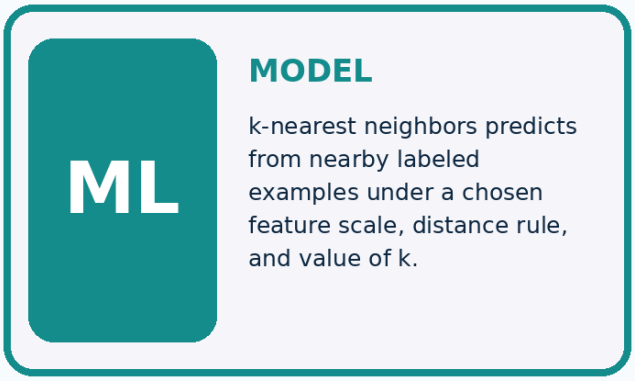
▼ Earlier wording, alternatives and picture transcripts

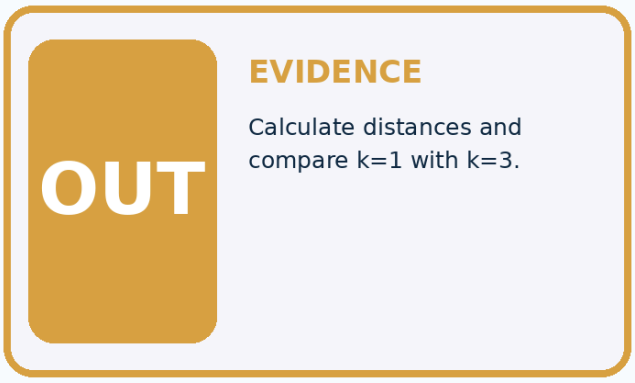
ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Calculate **distance**s and compare k=1 with k=3.

MODEL IDEA k-nearest **neighbor**s predicts from nearby labeled examples under a chosen feature scale, **distance** rule, and value of k.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD New leaf has normalized features (2, 3). Using Manhattan **distance**, add the absolute coordinate differences. **Training** leaves at (2, 2), (4, 3), and (1, 5) have **distance**s 1, 2, and 3 from (2, 3).

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Connect the picture and the explanation

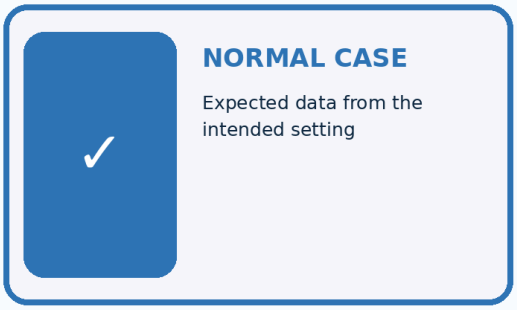
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

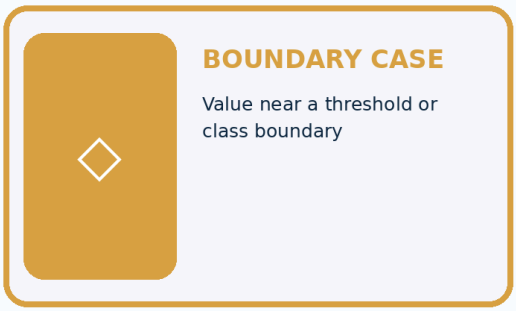
Compare numeric closeness with missing or changed feature evidence

> A neighbour vote is a method result, not verified truth.

Four evaluation conditions



NORMAL Expected data and expected behavior.

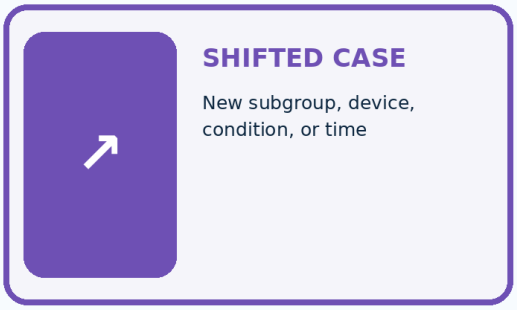


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: Move the query to length7 cm.

▼ **Check after predicting**

Prediction: k=1 predicts spotted from B, **distance**1; k=3 remains spotted.

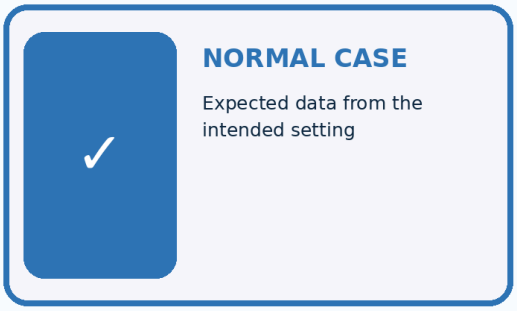
Reason: **Distance**s become 5, 1, 4 cm, so the nearest tie disappears.

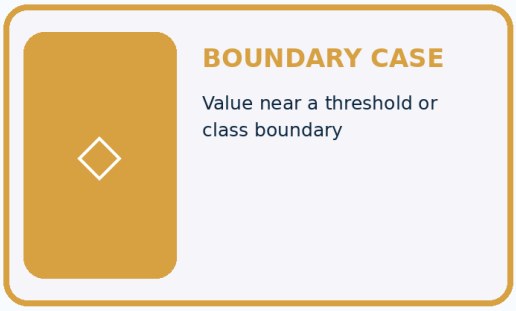
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

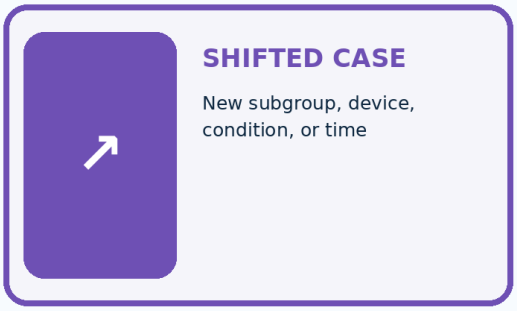
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Read each distance before voting

> Use the labelled number line to connect position with absolute difference.

Complete pictured points and scenario scope

Let a nearest-**neighbour** picture show each **distance** before voting.

Training: A length 2 healthy; B length8 spotted; C length11 spotted. Query length5 cm.
Distance is absolute difference; ties use alphabetically earlier ID.

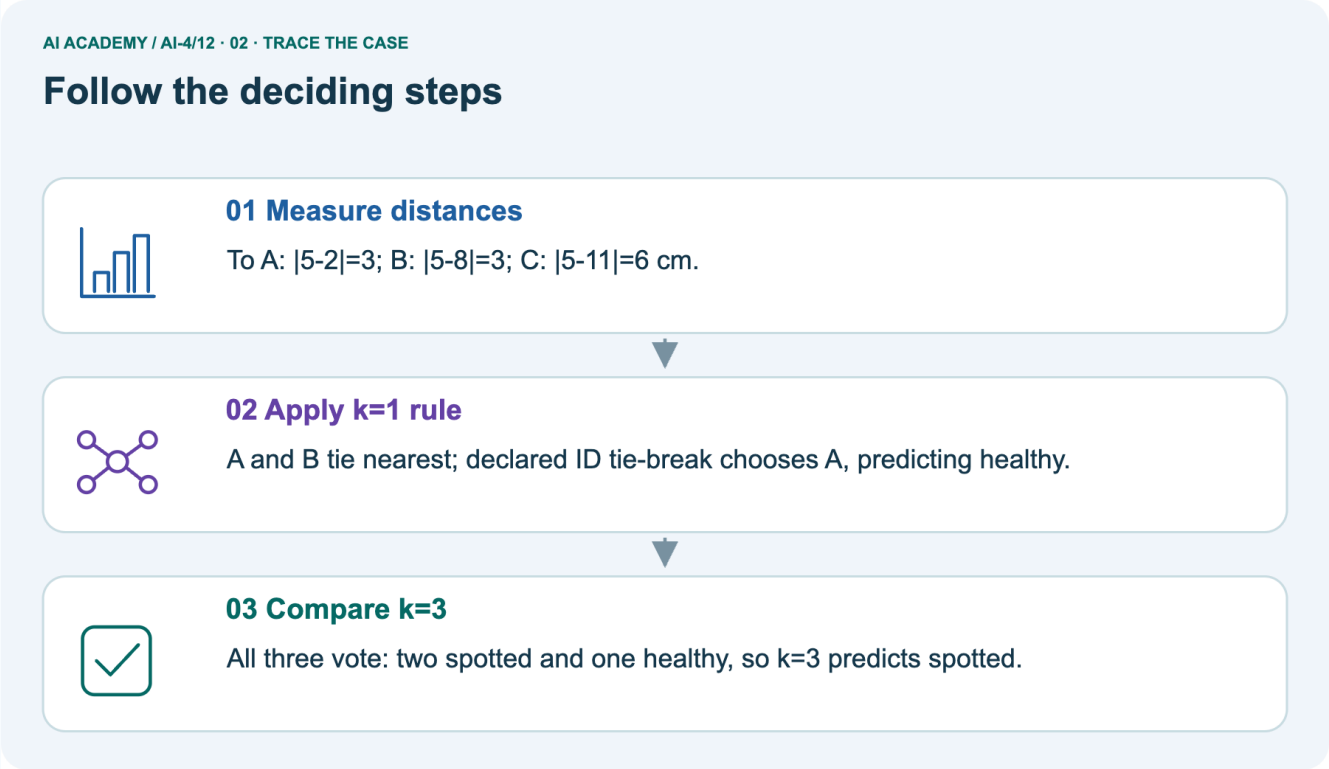
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Keep three examples separate. Main corridor query3 uses five stored readings A2 blocked,B4 clear,C8 clear,D10 blocked,E12 clear: k=1 chooses A blocked, k=3 chooses A/B/C clear. Picture uses only A2 healthy,B8 spotted,C11 spotted and query5: distances3,3,6; k=1 chooses

A healthy, k=3 spotted. Moving only that pictured query to7 gives distances5,1,4 and k=1 B spotted.

The plot’s vertical position is a fixed display row, not a second measurement. The independent task has its own three points and query7.

Trace distances, tie and vote



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Measure distances	To A: |5-2|=3; B: |5-8|=3; C: |5-11|=6 cm.
2. Apply k=1 rule	A and B tie nearest; declared ID tie-break chooses A, predicting healthy.
3. Compare k=3	All three vote: two spotted and one healthy, so k=3 predicts spotted.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

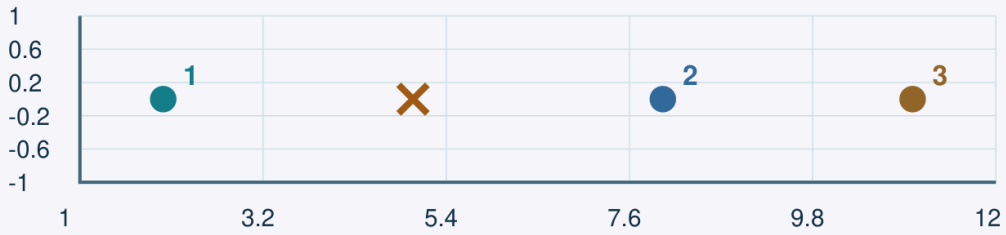
Read the picture in order. All three **vote**: two spotted and one healthy, so k=3 predicts spotted. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Inspect the original labelled length plot

AI ACADEMY / AI-4/12

One-dimensional neighbours shown along length axis



Horizontal: length (cm); vertical: fixed display row, y=0

1: A healthy at (2, 0)

2: B spotted at (8, 0)

3: C spotted at (11, 0)

Cross: query at (5, 0)

Distance rule: Absolute length difference; ties use alphabetically earlier ID

Follow the labelled picture with the matching explanation. Open the vector diagram

Predict moving only the pictured query

- Change one thing: Move the query to length7 cm. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting

AI ACADEMY / AI-4/12 · 04 · COMPARE THE CASES

Change one thing. Trace what follows.

Original case

All three vote: two spotted and one healthy, so k=3 predicts spotted.

Changed condition

Move the query to length7 cm.

Worked changed-case result

k=1 predicts spotted from B, distance1; k=3 remains spotted.
Why: Distances become 5, 1, 4 cm, so the nearest tie disappears.

Follow the labelled picture with the matching explanation. Open the vector diagram

State the feature and interpretation limits

- Length alone is a toy feature, not a disease diagnosis. **Distance** choices and feature **scaling** can change **neighbour**s.

- Read the labelled picture: Why is the k=1 outcome at length5 not uniquely determined without a tie rule? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to main corridor tie and stored-pool limits

Three selected neighbours for query 3

STORED EXAMPLE	FEATURE DISTANCE	ONE LABEL VOTE
A at 2 cm	abs 2 minus 3 is 1	blocked
B at 4 cm	abs 4 minus 3 is 1	clear
C at 8 cm	abs 8 minus 3 is 5	clear

Select neighbours by feature distance and the declared ID tie rule. Read their training labels to vote only after selecting them.

Store the **training** examples and the method settings. At **prediction** time, compare the query only with stored **training** examples. Do not add the query's reference label or other validation labels to the **neighbour** pool.

No cutoff or tree is fitted here, but retaining **training** examples is still part of the learning method.

Why do A and B tie for distance to query 3? Which wins for k=1 under our rule, and why does k=3 predict differently? Does the ID rule establish which label is true?

A **vote** share of two out of three describes these **neighbour**s. It is not a verified two-thirds probability that the new scene is clear. A **prediction** can be wrong even when the nearest examples agree.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

Four-step worked evidence for Week 12: Nearest Neighbors

STEP	EVIDENCE AND REASONING
1	New leaf has normalized features (2, 3).
2	Using Manhattan distance, add the absolute coordinate differences. Training leaves at (2, 2), (4, 3), and (1, 5) have distances 1, 2, and 3 from (2, 3).
3	With k=1, nearest label wins; with k=3, majority among all three wins.
4	Rescale features and retest because centimeters must not overwhelm a 0–1 color score.

Trace these corrected steps. Keep the stated metric, assumptions and interpretation when you change the case.

STEP 1 New leaf has normalized features (2, 3).

STEP 2 Using Manhattan **distance**, add the absolute coordinate differences. **Training** leaves at (2, 2), (4, 3), and (1, 5) have **distance**s 1, 2, and 3 from (2, 3).

STEP 3 With k=1, nearest label wins; with k=3, majority among all three wins.

STEP 4 Rescale features and retest because centimeters must not overwhelm a 0–1 color score.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

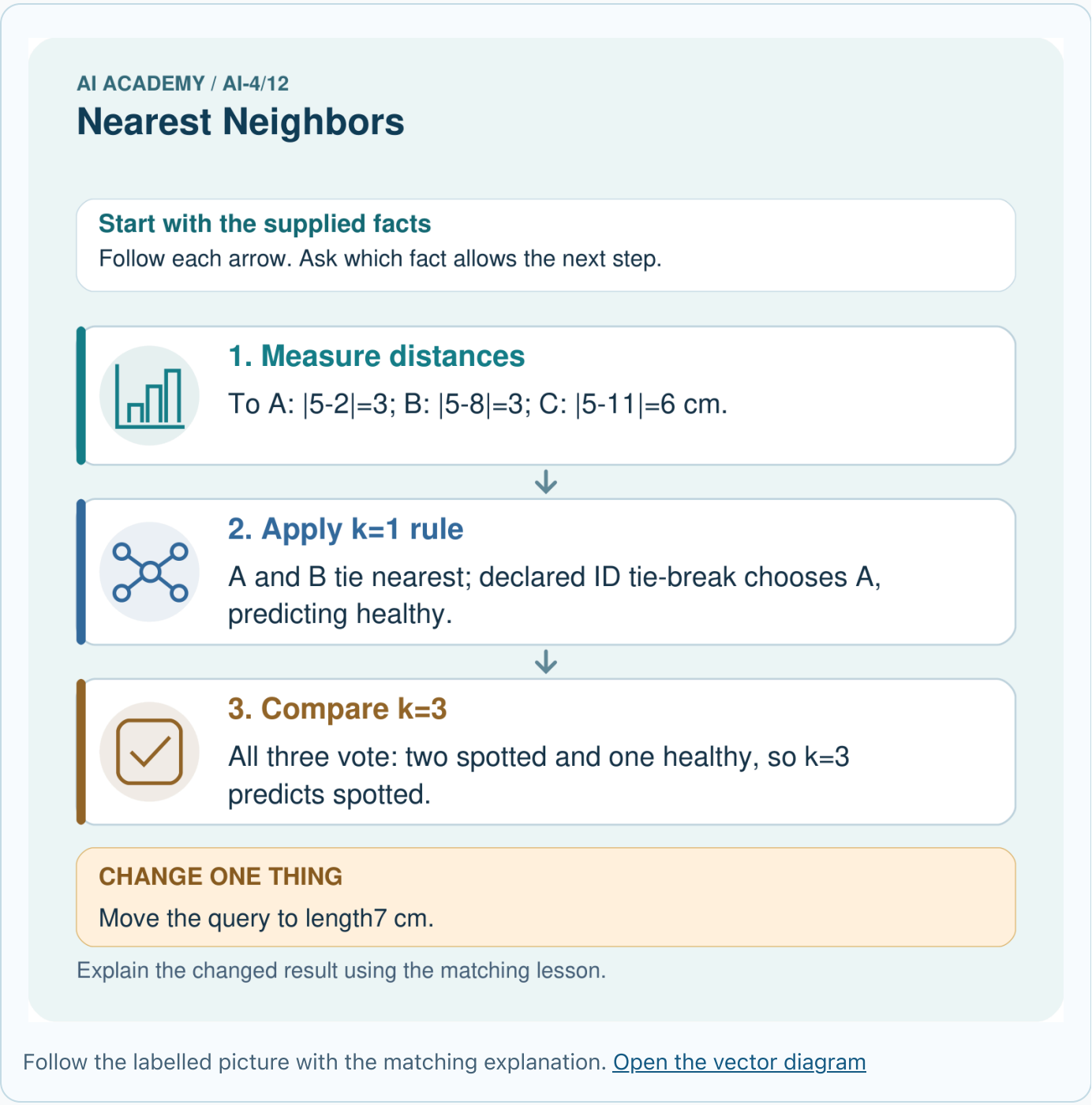
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Nearest Neighbors

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find the first unsupported distance or probability claim

> Negative coordinate differences become nonnegative absolute distances.

Diagnose the main claim

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE Nearest **neighbor** means the photo that looks most similar to a person.

Explain the method and remaining uncertainty

- All **distance**s here use one feature measured in the same unit. With multiple features, their units and scales affect which examples appear close. A large numerical range can dominate a **distance** calculation.

  Next week we will address **scaling** and learn any **scaling** values from **training** data only.

- Identical readings from different trips can have different labels, and an exact match does not certify truth. Keep related examples grouped across splits as in Week 6.

  Testing an example against a pool containing itself can make a one-**neighbour** score misleadingly high; it does not show performance on a new trip.

> **A learner adds each labelled validation query to the stored pool, then uses k=1 and reports perfect predictions. Explain what the result fails to test and how to restore the intended comparison.**

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: Nearest neighbor means the photo that looks most similar to a person. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

A closer explanation

Practise fixed-k predictions and their actual validation errors

> Keep validation examples outside the neighbour pool.

Try W2 with the complete training table

Use workbook W2 for the connected example “Nearest **neighbour**s and **distance**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Complete three-query validation and baseline comparison

Keep k=3 and the original five **training** rows. Validation queries with reference labels are (3,blocked), (9,blocked), (11,clear). Their selected **neighbour** IDs are A B C, C D E, and D E C respectively. Each group **vote**s clear.

Only the last **prediction** matches, so accuracy is 1/3, about 33.3 percent.

The **training**-majority baseline also predicts clear because three **training** labels are clear and two are blocked. It gets the same 1/3 on these validation rows. Record all **distance**s, selected IDs and errors.

Trying more k values is a development choice to document; preserve a separate final test for a final evaluation.

Challenge negative-distance reasoning

- Use the worked case for Nearest **neighbour**s and **distance**s. Proposed explanation: “A negative difference is a negative **distance**”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the pictured tie rule

- Why is the k=1 outcome at length5 not uniquely determined without a tie rule? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Reserve the independent three-point task

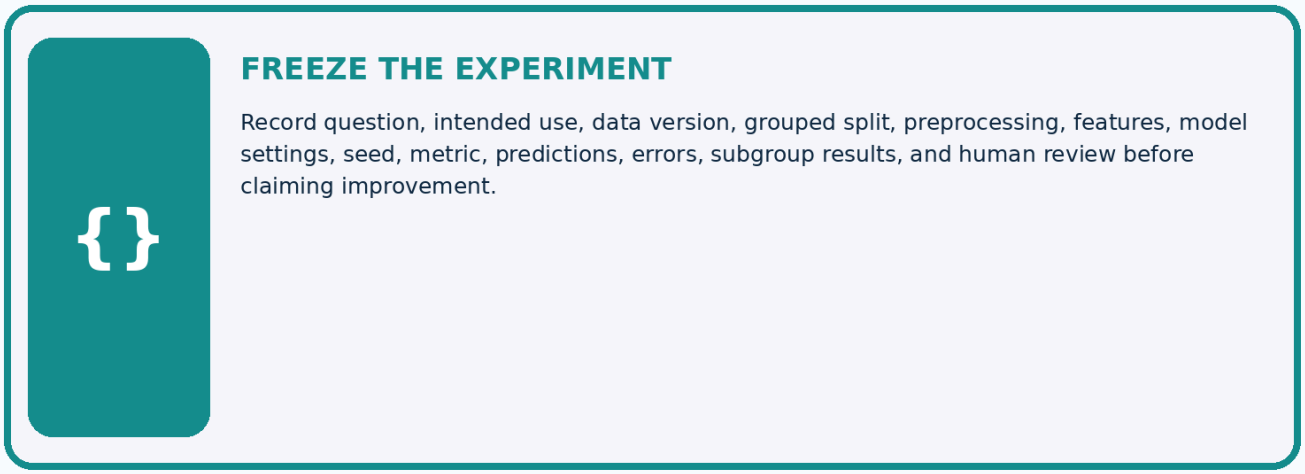
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Practise with less help: Apply neighbour and tie rules

Check your reasoning

Explain the picture

Run a complete distance map and validation comparison

> The supplied rows support every rank, vote and denominator.

Complete three-query distance and neighbour table

Main validation query / reference	Distances to A,B,C,D,E	k=3 nearest IDs / prediction
3 / blocked	1,1,5,7,9	A B C / clear
9 / blocked	7,5,1,1,3	C D E / clear
11 / clear	9,7,3,1,1	D E C / clear

Normal, exact-match, missing and far-query probes

Complete neighbour probe	Expected bounded check
Normal main query3,k=3	A/B/C vote clear.
Exact match query8,k=1	Stored C is distance0 and predicts clear; a matched training feature is not proof of query truth.
Missing query reading	Original program assumes numeric input; request evidence/review rather than substituting0.
Shifted main query100,k=3	Distances98,96,92,90,88; select E,D,C → clear, despite being far beyond stored readings. Distance-based prediction alone does not establish reliability.

Measure, rank, vote and compare

1. Neighbour Map: compute all five distances for each complete main validation row, sort by distance/ID, and compare k=1 with the predeclared k=3. For the fixed k=3 method and training-majority clear baseline, score1/3 each; record wrong rows rather than claim superiority. Use separate tables for the three-point picture and optional two-feature scaling example.

   Keep the pool training-only and save units, feature order, distance, k and tie policy. No model probabilities, new measurements or real lane-safety results are supplied. Keep the independent query7 task for an uncoached first attempt.

▼ Optional complete scaling contrast

Complete optional two-feature example	Supplied numeric evidence
Training features available before prediction	A(10cm,200 brightness units) clear; B(20,100) blocked; C(20,200) blocked. Query(11,120). k=1; Manhattan distance; ties alphabetical ID.
Raw distance sums A/B/C	81,29,89 → B blocked. This adds raw coordinate differences under a declared numerical rule.
Training-only scaling	Feature minima(10,100), ranges(10,100); transform each coordinate as(value-minimum)/range.
Scaled training/query	A(0,1),B(1,0),C(1,1); query(0.1,0.2).
Scaled distances A/B/C	0.9,1.1,1.7 → A clear. Feature contributions changed, labels unchanged.
Limits	Do not refit scale on the query or validation; units/scaling choice does not prove real feature importance or better accuracy.

Keep the evidence and actual execution status

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____

> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

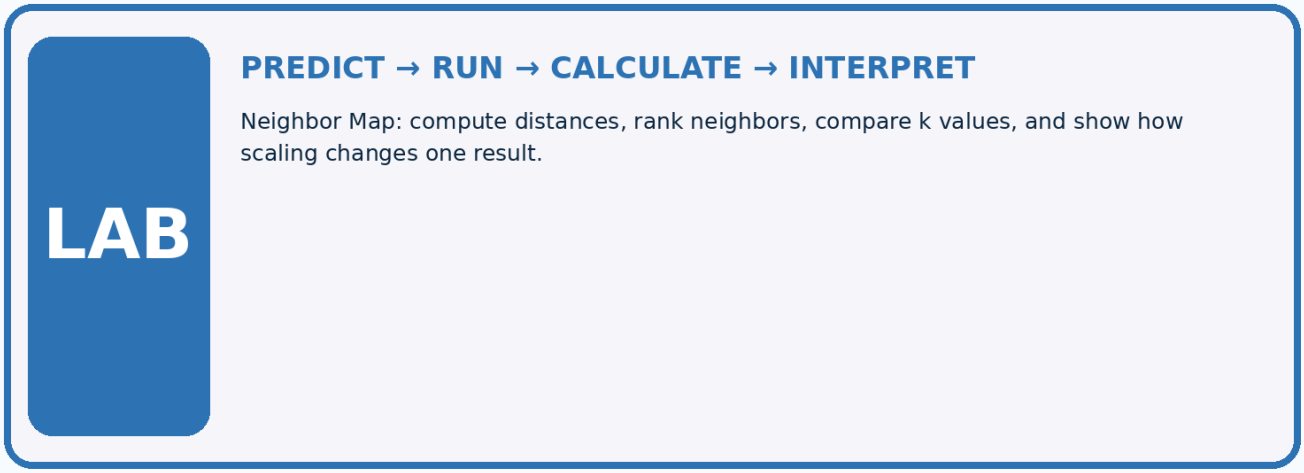
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. **Neighbor** Map: compute **distance**s, rank **neighbor**s, compare k values, and show how **scaling** changes one result.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: compare k=1 and k=3 on new points

> The tie rule selects a prediction without establishing its truth.

Complete selected task and scope

Training points: A length1 healthy, B length5 spotted, C length9 healthy. Query length7. Use absolute **distance**; for a nearest-**neighbor** tie choose the alphabetically earlier ID. Compare k=1 and k=3.

Use only this selected task's A/B/C training points and query7. Rank by absolute length difference, then alphabetical ID on equal distance; select one or all three as requested. No reference label for the query is supplied, so report method predictions rather than accuracy or proof of health. Do not import the picture's point labels or the main five-row corridor pool.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Changed independent case

Week 12 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

For one feature, labeled points are 2:fern and 8:grass. A new point is 6. Under one-nearest-**neighbor** with absolute **distance**, predict its label.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Implement a transparent k-NN paper prototype for the Plant Classifier and record the chosen features, **scaling**, **distance**, and k.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

Training lengths 2(H),6(S),10(H); query=8. Compute **distance**s. Explain the tie and why combining length in mm with width in cm changes a two-feature **distance**.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the neighbour method reproducible with the project

> Another learner should be able to rebuild the same ranked list.

Project artifact and evidence

THIS WEEK'S CHECKPOINT Implement a transparent k-NN paper prototype for the Plant Classifier and record the chosen features, **scaling**, **distance**, and k.

Keep the source-separated stored training examples, feature names/units/order, any training-fitted scale, distance, k and tie rules. Save per-query distances, selected IDs, predictions and same-row baseline errors. Handle missing readings under a declared review policy. Report distant queries and sparse evidence; a2/3 neighbour vote is not automatically a calibrated probability.

Development selection of k requires a separate later final evaluation.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Limits and human decision

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

▼ **Earlier wording, alternatives and picture transcripts**

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.

RESPONSIBLE PLANT CLASSIFIER

Implement a transparent k-NN paper prototype for the Plant Classifier and record the chosen features, scaling, distance, and k.

Evidence for your project

MAIN LESSON · SECTION 16

Finish with the nearest IDs and the rule that selected them

> Explain what a neighbour vote can and cannot establish.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Compute absolute **distance**s and rank **training neighbour**s

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
distance	numeric measure of dissimilarity	
neighbor	nearby labeled training example	
k	number of neighbors voting	
scaling	make feature ranges comparable	

Closer-explanation terms

Word	Meaning
Distance	A numerical measure of difference between selected feature values
Neighbour	A stored training example close under the chosen distance
k	The number of neighbours selected for a prediction

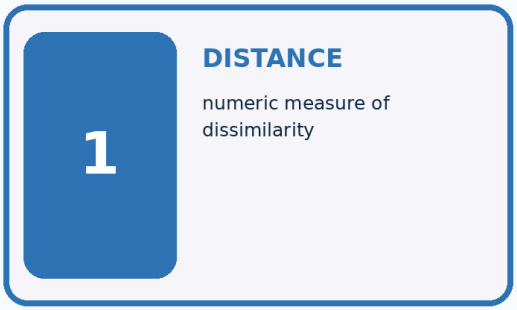
Word	Meaning
Vote	A contribution of a selected neighbour’s label to the predicted class

▼ Earlier wording, alternatives and picture transcripts

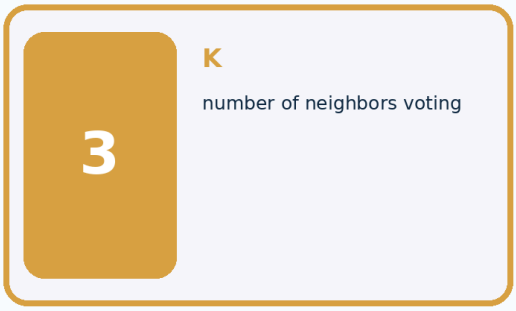
VOCABULARY

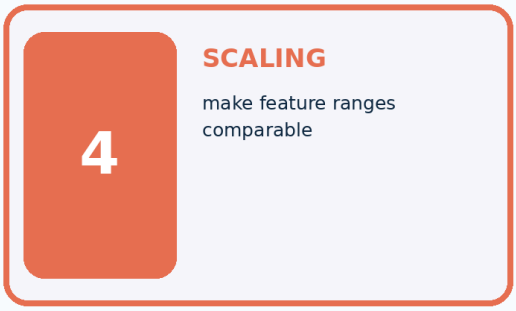
Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR Nearest **neighbor** means the photo that looks most similar to a person.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **distance**, **neighbor**, k, **scaling**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE Nearest **neighbor** means the photo that looks most similar to a person.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

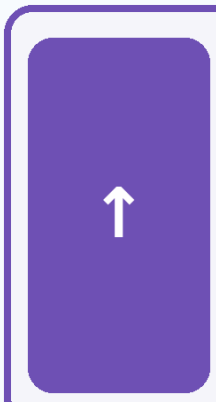
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



ML BUILDER MASTERY

1 Define • 2 Calculate • 3 Trace code • 4 Run experiment • 5 Diagnose errors • 6 Improve one cause • 7 Defend limits

▼ **Reference · Inspect your evidence**

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the selected new three-point query7 neighbour task for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Calculate **distance**s and compare k=1 with k=3.

NEXT BRIDGE Carry the experiment **evidence** into: Train a Classifier.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

One-dimensional **training** points: 2 healthy, 4 healthy, 8 spotted. Query=6; absolute **distance**.

Find k=1 **neighbour**s and handle ties.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

Distances 4,2,2. Points 4 and 8 tie with different labels. Use a predefined tie policy or uncertainty. **Scaling** matters with differently measured features.

Evidence or stage	Value or result
Point 2, healthy	4
Point 4, healthy	2
Point 8, spotted	2

▼ Earlier wording, alternatives and picture transcripts

Week 12 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Absolute distance from query 6

One-dimensional **training** points: 2 healthy, 4 healthy, 8 spotted. Query=6; absolute **distance**.

Find k=1 **neighbour**s and handle ties.

Distances 4,2,2. Points 4 and 8 tie with different labels. Use a predefined tie policy or uncertainty. **Scaling** matters with differently measured features.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Compute absolute distances and rank training neighbours	Explain the deciding step, show a trace or test, and state one limit.
Apply a declared neighbour count and distance tie rule	Explain the deciding step, show a trace or test, and state one limit.
Evaluate neighbour predictions without using validation labels as neighbours	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Let a nearest-**neighbour** picture show each **distance** before voting.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Let a nearest-**neighbour** picture show each **distance** before voting.

- Which labelled step matches this detail? All three **vote**: two spotted and one healthy, so k=3 predicts spotted.
- 1. Measure **distance**s
- 2. Compare k=3
- 3. Apply k=1 rule
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Compare k=3 All three **vote**: two spotted and one healthy, so k=3 predicts spotted. Read the picture in order.

All three **vote**: two spotted and one healthy, so k=3 predicts spotted. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Move the query to length7 cm. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.

- 2. k=1 predicts spotted from B, **distance**1; k=3 remains spotted.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: k=1 predicts spotted from B, **distance**1; k=3 remains spotted. **Distance**s become 5, 1, 4 cm, so the nearest tie disappears.

Repair and keep the record

Review Decision Trees (ai-4/11). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Which **input** distinguishes the original and changed tree?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Dataset Cards (ai-4/9). Goal: Inspect row meaning missing values and repeated records. Short practice: What two counts should the revised card show?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Label Quality (ai-4/6). Goal: Write an observable label rule with boundaries and a review option. Short practice: Why should two annotators not **vote** away Z's missing **evidence**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work
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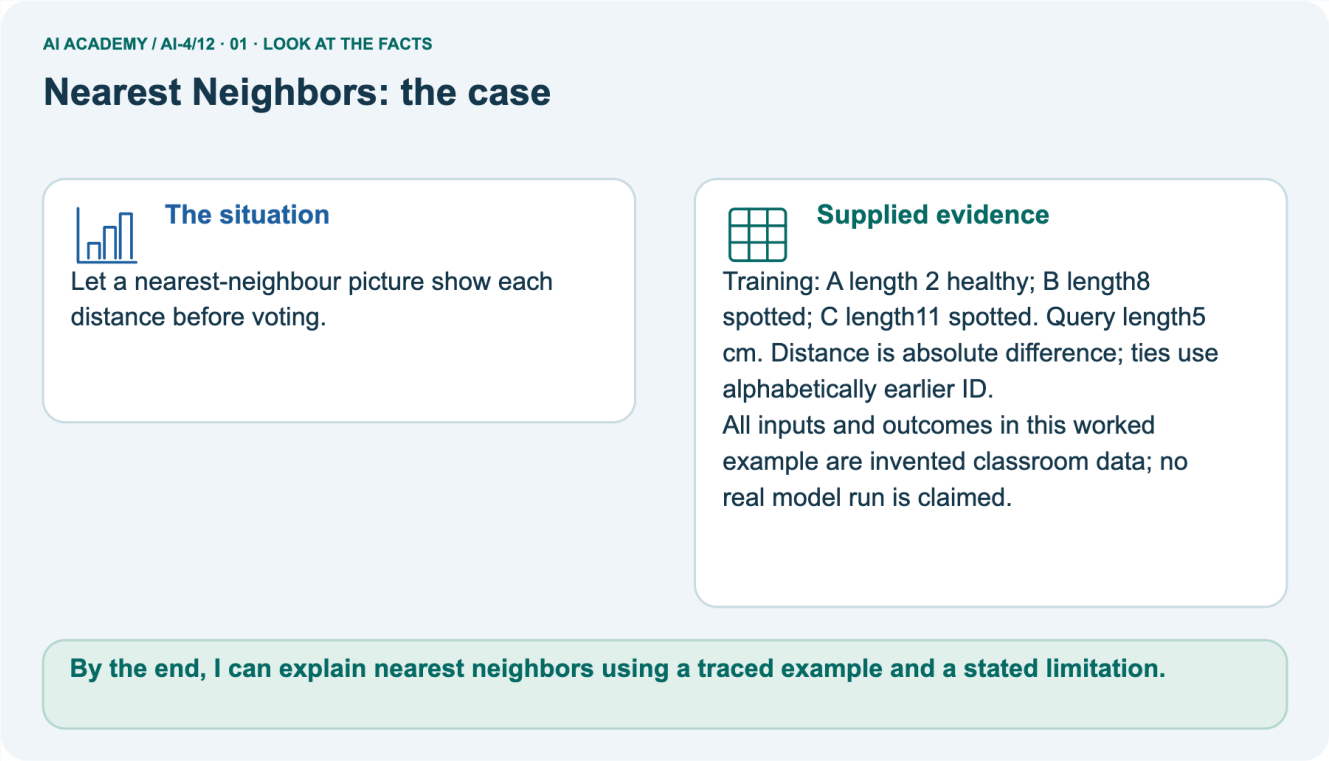
<details>
<summary>Reference · The six-picture journey</summary>
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REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Nearest Neighbors: the case



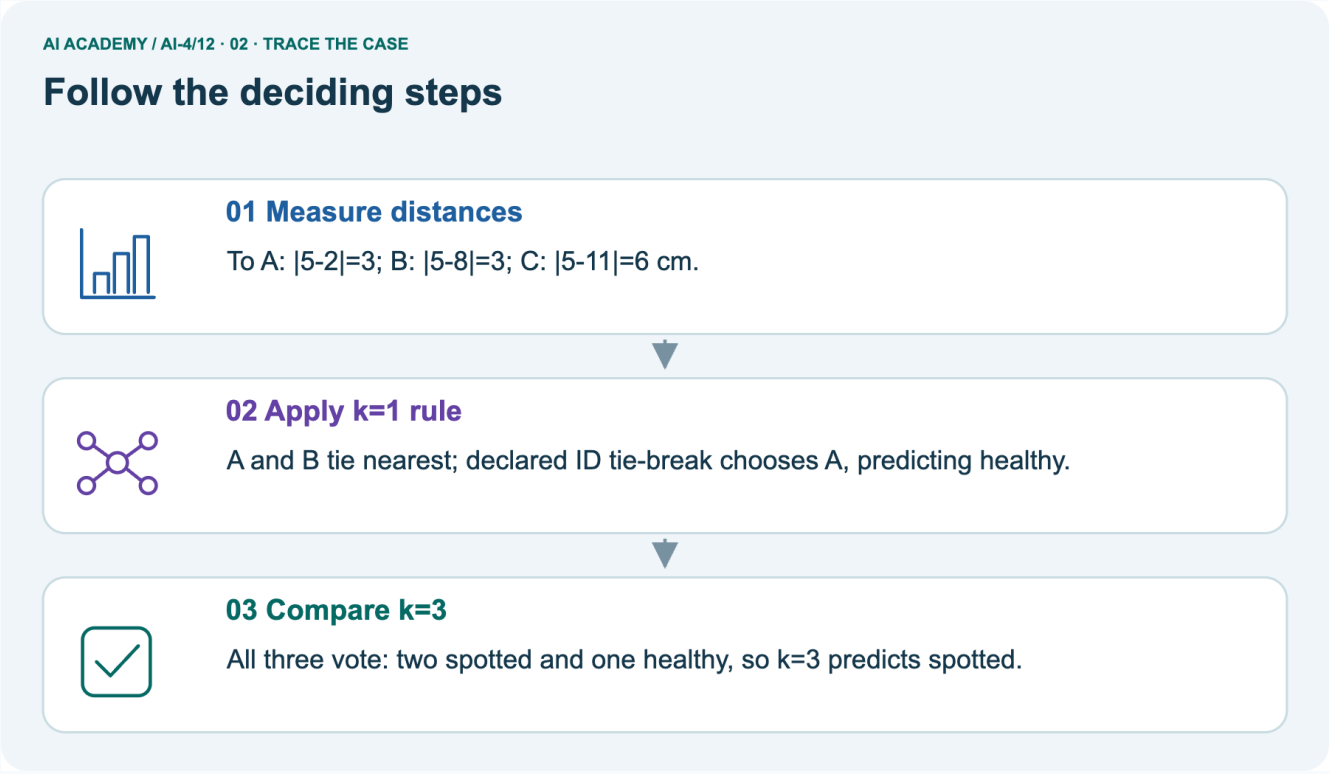
Follow the labelled picture with the matching explanation. Open the vector diagram

Let a nearest-**neighbour** picture show each **distance** before voting. Supplied facts: **Training**: A length 2 healthy; B length8 spotted; C length11 spotted. Query length5 cm.

Distance is absolute difference; ties use alphabetically earlier ID. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

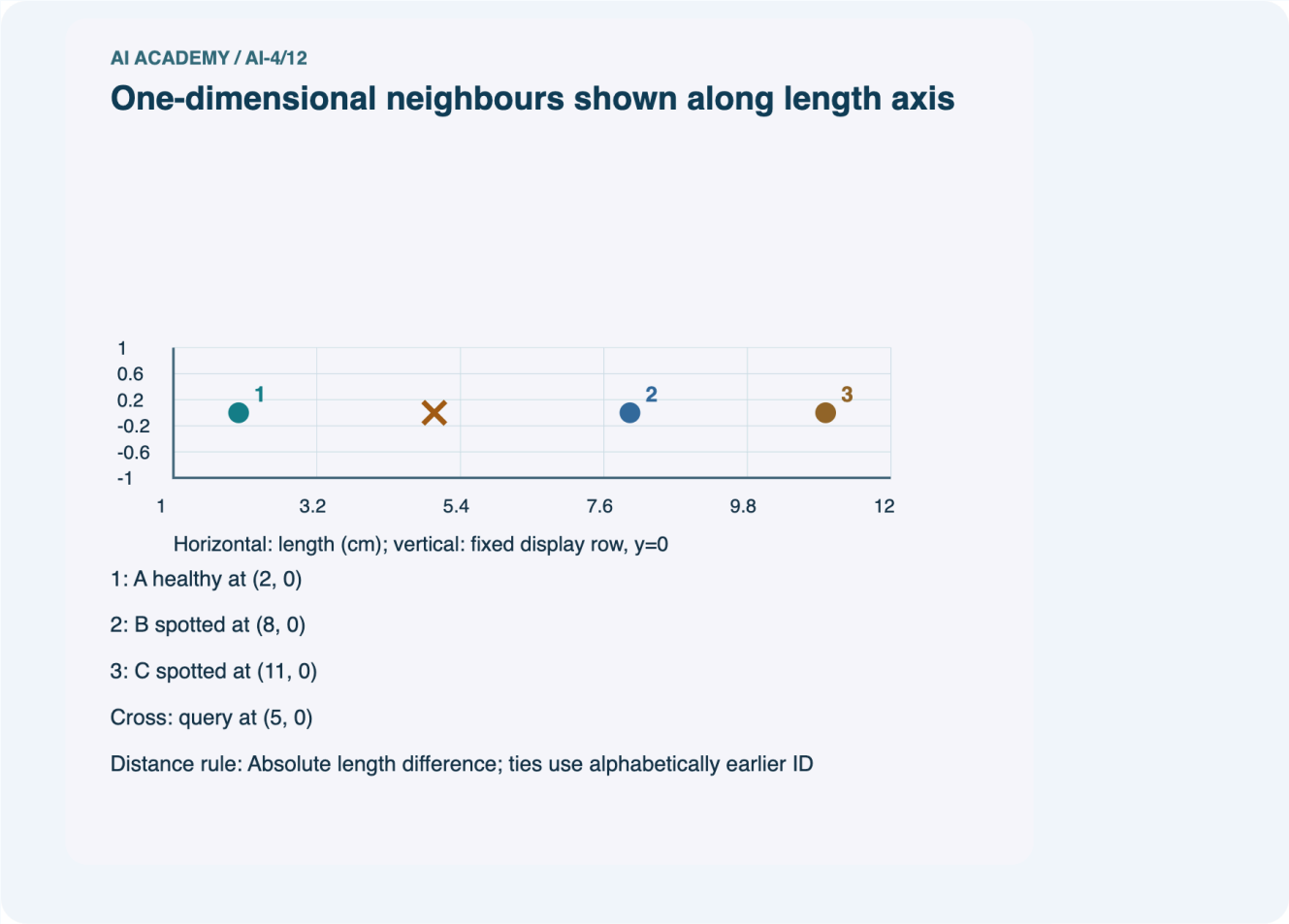


Follow the labelled picture with the matching explanation. Open the vector diagram

Measure **distance**s: To A: |5-2|=3; B: |5-8|=3; C: |5-11|=6 cm. → Apply k=1 rule: A and B tie nearest; declared ID tie-break chooses A, predicting healthy. → Compare k=3: All three **vote**: two spotted and one healthy, so k=3 predicts spotted.

Point to each numbered stage and explain which recorded **input** it uses.

03. One-dimensional neighbours shown along length axis



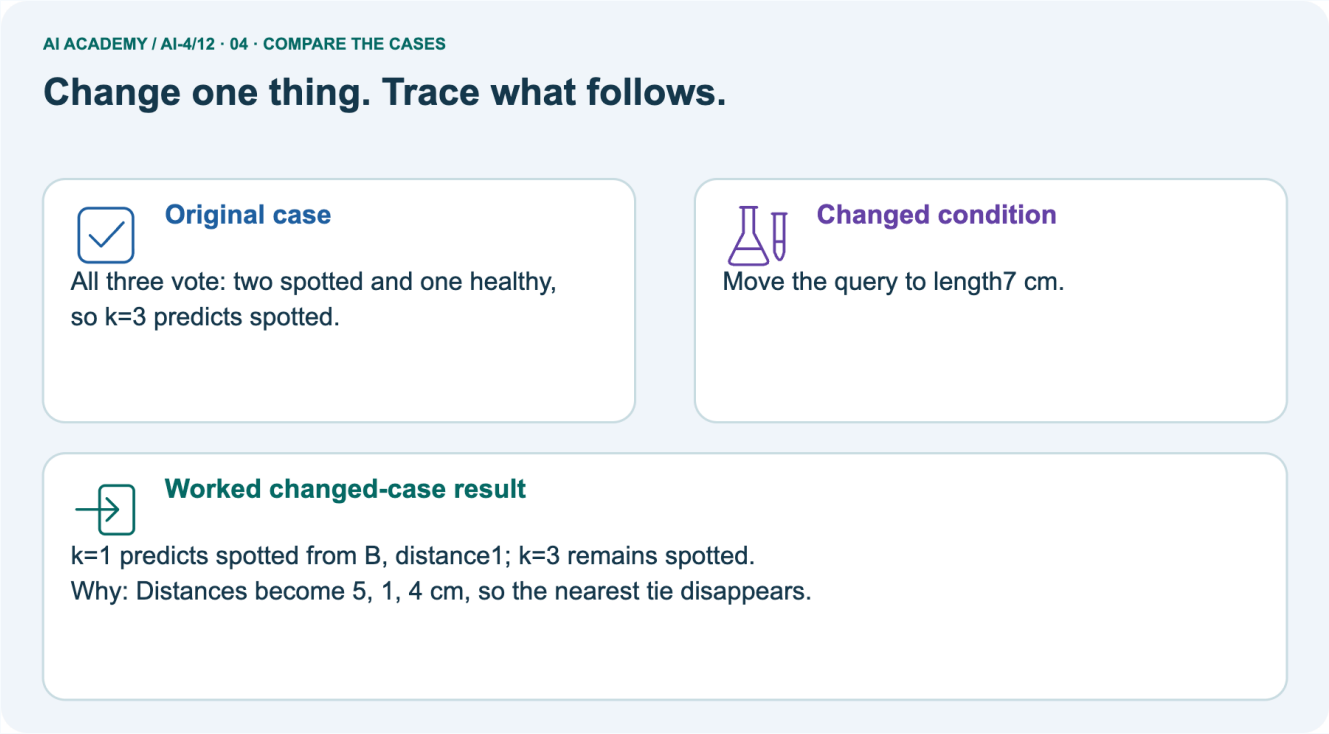
Follow the labelled picture with the matching explanation. Open the vector diagram

One-dimensional **neighbour**s shown along length axis. A healthy: (2, 0). B spotted: (8, 0). C spotted: (11, 0). Query query: (5, 0). Axes: length (cm); fixed display row, y=0.

Distance: Absolute length difference; ties use alphabetically earlier ID.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



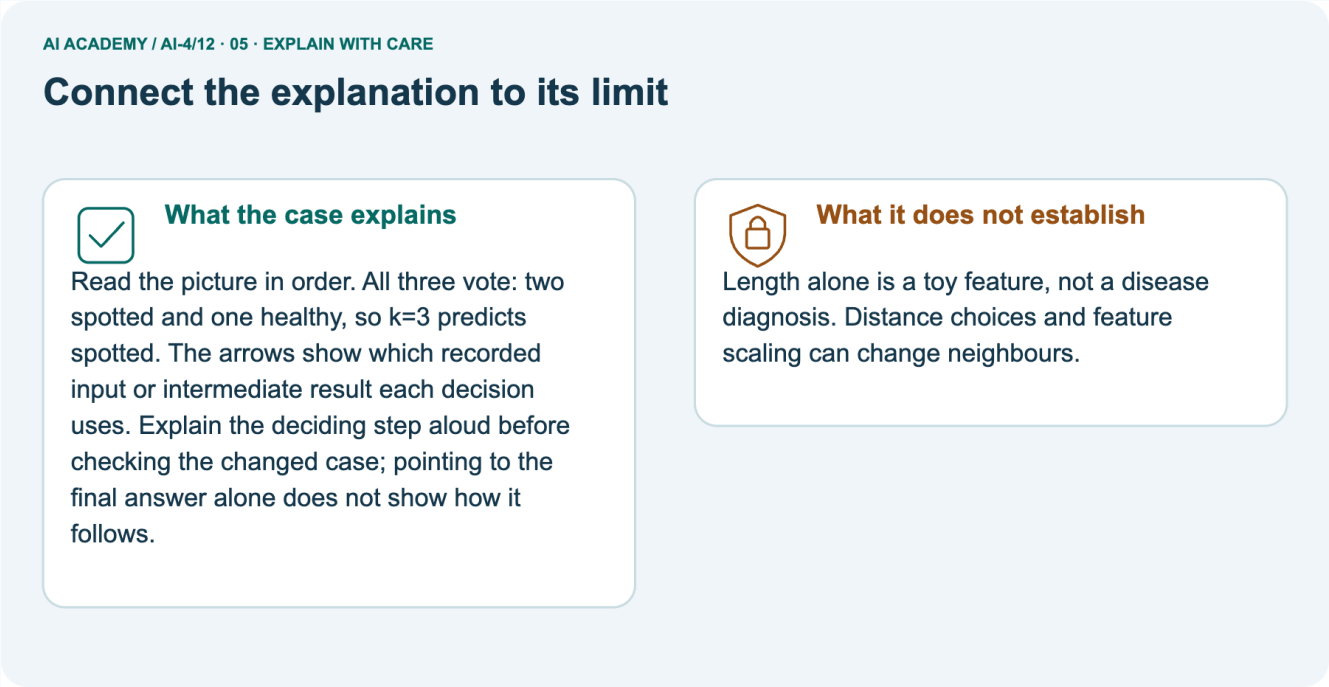
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: All three **vote**: two spotted and one healthy, so k=3 predicts spotted. Changed condition: Move the query to length7 cm. Expected result: k=1 predicts spotted from B, **distance**1; k=3 remains spotted.

Reason: **Distance**s become 5, 1, 4 cm, so the nearest tie disappears.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

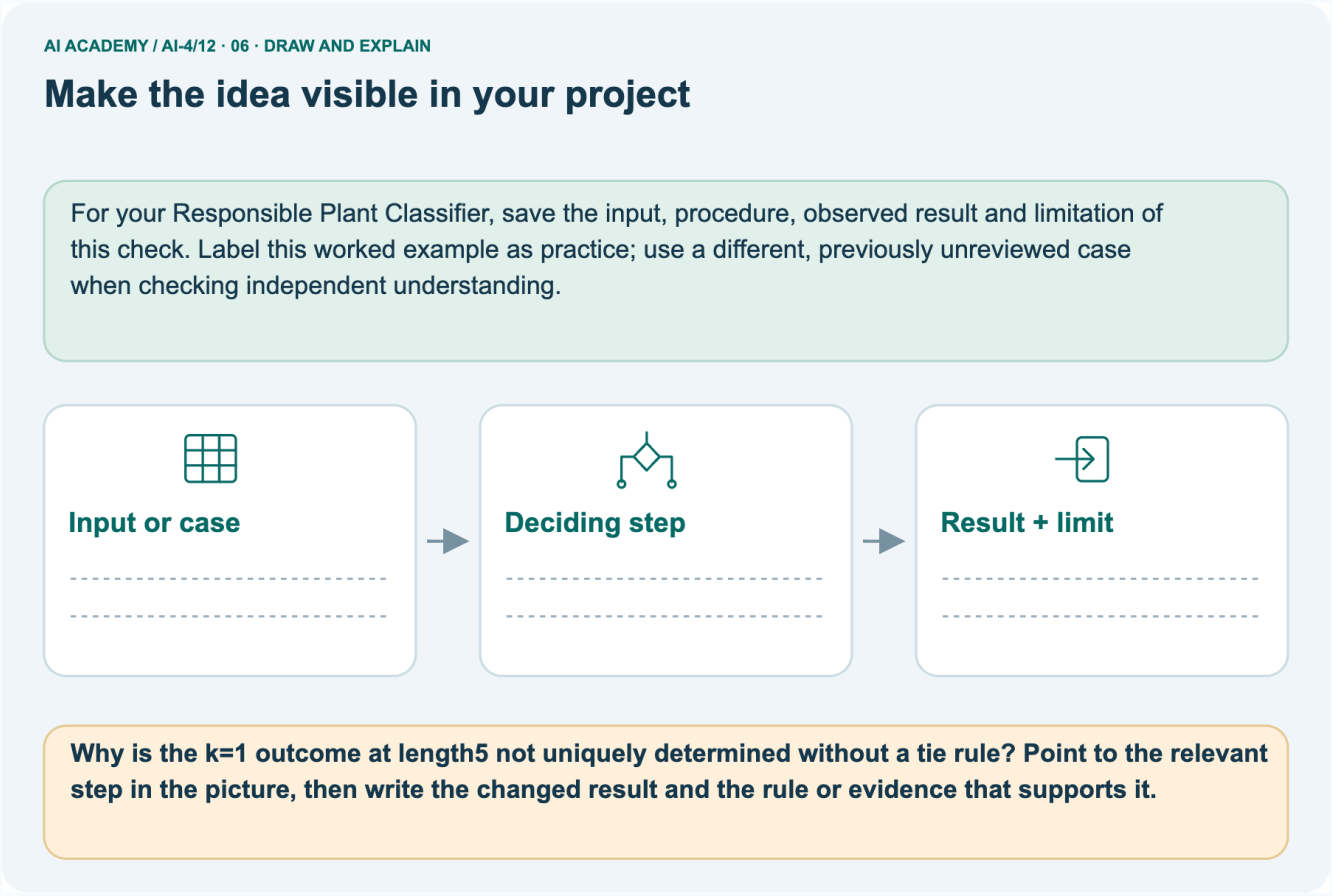
Read the picture in order. All three **vote**: two spotted and one healthy, so k=3 predicts spotted. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Length alone is a toy feature, not a disease diagnosis.

Distance choices and feature **scaling** can change **neighbour**s.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is the k=1 outcome at length5 not uniquely determined without a tie rule? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Nearest Neighbors: the case

02. Follow the deciding steps

03. One-dimensional neighbours shown along length axis

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.